

TFScope: sequence-only prediction of transcription-factor binding specificity rivals structure-based methods

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Abstract

Predicting the DNA sequence preference of a transcription factor (TF) from its protein sequence alone remains a central problem in gene-regulatory genomics. State-of-the-art accuracy is achieved by structure-based models that require an experimentally determined or predicted protein–DNA complex, limiting their use for the large fraction of TFs and designed proteins that lack a reliable structure. Here we introduce **TFScope**, a sequence-only model that maps a TF DNA-binding domain (DBD) to a position weight matrix (PWM) using a *frozen* protein language model (ESM-2 650M) adapted with low-rank (LoRA) fine-tuning, a residue mixture-of-experts, and a contact-aware motif head. On a leakage-controlled benchmark in which held-out TFs share no training identity, a five-seed TFScope ensemble reaches a motif content correlation of 0.664 and is statistically indistinguishable from the structure-based method DeepPBS (mean gate-oracle r 0.657 vs 0.626; bootstrap $p = 0.30$), despite using no structure at inference. An independent structure predictor (Boltz-2) judges TFScope’s sequence-only consensus motifs to form protein–DNA complexes at least as confidently as DeepPBS’s (ipTM 0.957 vs 0.923; 32/41 TFs; Wilcoxon $p = 1 \times 10^{-5}$). Probing the frozen backbone shows that it linearly encodes DNA-contact residues (AUROC 0.95) and organizes DBD families (k-NN purity 0.83), explaining where the signal comes from; a variance decomposition attributes 59% of TFScope’s edge to genuine within-family specificity rather than family lookup. We further show directional single-mutation motif switching (MyoD1 E-box) and application to *de novo* designed DBDs, and we transparently characterize the model’s principal limitation — insensitivity to specificity-altering point mutations inherited from the frozen backbone. TFScope makes accurate specificity prediction available wherever only a protein sequence is known.

Introduction

Transcription factors (TFs) interpret the genome by binding short, degenerate DNA motifs, and the position weight matrix (PWM) that summarizes this preference is one of the most widely used objects in regulatory genomics. PWMs drive motif enrichment analysis, the annotation of *cis*-regulatory elements, and the interpretation of non-coding variants, and they anchor the priors used by many chromatin and gene-expression models^{1–3}. A PWM is therefore not merely a descriptive statistic but a reusable input to a large downstream ecosystem, which makes the availability of accurate PWMs a practical bottleneck rather than an academic curiosity.

Yet measured specificities remain sparse. High-throughput assays such as protein-binding microarrays and HT-SELEX have characterized many TFs, but the experiments are laborious and have covered only a fraction of the roughly 1,600 human TFs, with even thinner coverage across non-model organisms^{1,4}. The gap is widening rather than closing, because protein design now produces *de novo* DNA-binding proteins far faster than any binding assay can profile them. A model that predicts a PWM directly from a protein sequence would let this downstream ecosystem operate wherever a sequence is known, including the majority of TFs and essentially all designed proteins that will never be assayed.

Computational specificity prediction has followed two routes with complementary weaknesses. Structure-based models read the physicochemical readout of a protein–DNA complex and currently define the state of the art; the geometric deep-learning method DeepPBS, for example, predicts specificity from the geometry and chemistry of the interface⁵. Their accuracy, however, is gated on the availability and quality of a structure. For a TF without a co-crystal, and for any designed protein, a complex must first be predicted, which adds computational cost and a second, compounding source of error. Sequence-only models avoid this dependency but have historically been less accurate, and their reported advantages are frequently inflated by TF leakage, in which test proteins share close homologs with the training set and the model succeeds by recall rather than by genuine generalization⁵. Distinguishing real sequence-to-specificity inference from leakage is thus as important as raw accuracy.

Protein language models (PLMs) trained on evolutionary-scale sequence data have reshaped what is achievable from sequence alone, recovering contacts, structure, and function that were previously thought to require experimental input^{6,7}. We reasoned that a frozen PLM might already encode the recognition information that specifies DNA binding, and that the remaining problem is not representation learning but read-out: extracting a calibrated motif from features the backbone has already computed. This framing predicts that a small, structure-free adapter on a frozen backbone should suffice, and that the ceiling on accuracy should be set by the backbone rather than by the adapter.

Here we test this hypothesis with **TFScope**, a sequence-only model that maps a TF DNA-binding domain to a PWM through a frozen ESM-2 650M backbone and a lightweight contact-aware adapter. Under strict leakage control, TFScope reaches accuracy parity with the structure-based method DeepPBS without ever seeing a structure at inference. An independent structure predictor, blind to both models, judges TFScope’s sequence-only motifs to form protein–DNA complexes at least as confident as those from DeepPBS. Probing the frozen backbone shows why this is possible and where the signal originates, a variance decomposition establishes that the model reads specificity within TF families rather than merely retrieving a family average, and we characterize transparently the one regime, single specificity-altering point mutations, where a frozen backbone remains limited.

Results

A structure-free model of transcription-factor specificity

TFScope maps a TF DNA-binding domain (DBD) to a $4 \times L$ PWM in a single forward pass, using no structural input at any stage of inference (Fig. 1). The design follows directly from the read-out hypothesis: because we assume the recognition information already exists in the backbone, we keep the backbone frozen and invest the model’s limited capacity in extracting and calibrating a motif from it. Concretely, per-residue features are taken from a frozen ESM-2 650M encoder⁷ as a learned weighted average of its last four transformer layers, which lets the model draw on both

the syntactic features of late layers and the more local features of earlier ones, and each residue is tagged with a binary indicator of whether it lies inside the annotated DBD.

Four lightweight components then convert these frozen features into a PWM. Low-rank adaptation⁸ (LoRA, rank 16) supplies the only updates that touch the backbone, adapting attention projections at a small parameter cost while leaving the pretrained representation intact. A token-level residue mixture-of-experts, with two shared and eight routed experts conditioned on a TF family embedding, allows different structural classes of DBD to be processed by specialized parameters without fragmenting the model into per-family heads. Dual gated-attention pooling then summarizes the residues twice, once globally and once restricted to the DBD, so that the motif read-out sees both the domain context and the recognition surface. Finally, a position gate head emits a soft gate over motif positions together with a pooled protein representation, and a contact-aware PWM head predicts a prior PWM and adds a learned correction, so that the final motif is the sum of a coarse prior and a contact-informed refinement. The correction is the only place structural knowledge enters the model, and it does so only during training, through supervision from recognition-residue DNA contacts; at inference the head runs from sequence alone.

TFScope is trained on curated protein-to-PWM pairs using DBD-cropped inputs, so that the model learns to read the recognition domain rather than incidental flanking sequence. Because single models are sensitive to the arbitrary register in which a short motif is committed, we report a five-seed ensemble whose members are aligned to a common reference frame by offset and reverse-complement before averaging, which removes register disagreement and yields a more stable motif than any single seed. Full architecture, training, and ensembling details are given in Methods.

TFScope enables specificity prediction without structure

To ask whether sequence-only prediction can rival a structure-based model, we benchmarked TFScope against DeepPBS⁵ on a shared held-out set (Fig. 4). The comparison is only meaningful if leakage is controlled, because naive splits reward a model for recognizing a training homolog rather than for generalizing to a new protein. We therefore imposed two independent controls. First, we used a held-out set clustered at 40% DBD identity, so that test proteins are not close homologs of any training protein. Second, for the direct head-to-head we retrained DeepPBS after removing the benchmark genes from its training data, producing a gene-disjoint comparison in which neither model has seen the test TFs. Throughout, we scored PWMs with a register-invariant, oracle-aligned motif-content correlation that permits an offset and reverse-complement before comparison, the appropriate metric for methods that do not share DeepPBS’s fixed crystallographic coordinate frame; scoring in a fixed frame would penalize a sequence-only method for register alone rather than for the motif it predicts.

On the clustered holdout (84 structures spanning 41 TFs), the five-seed TFScope ensemble reached a motif-content correlation of **0.664**, ahead of the single-seed model (0.629) and of every intermediate configuration in an ablation ladder that adds the model’s components one at a time (Fig. 4). Against DeepPBS under the matched per-motif protocol, TFScope scored 0.657 and DeepPBS 0.626, a paired difference of +0.031 that a bootstrap test does not distinguish from zero ($p = 0.30$). The two methods are therefore statistically indistinguishable, with the point estimate favoring the sequence-only model. The gene-disjoint retrain gave the same conclusion from the opposite direction: the retrained DeepPBS ensemble scored 0.720 and TFScope 0.685 on those 20 genes, a gap of 0.034 whose confidence interval crosses zero. We note that the frequently quoted pretrained-DeepPBS value of 0.806 is not a valid comparison here, because those checkpoints were trained on all 20 test genes and the number reflects leakage rather than generalization. Taken together, and consistent across two orthogonal leakage controls, TFScope attains structure-level

accuracy while using no structure at inference. Per-family performance (Fig. 3) shows that parity is broad rather than driven by a single easy family, spanning homeodomain, bHLH, bZIP, ETS, C2H2 zinc-finger, and nuclear-receptor DBDs, and representative predicted logos recover the canonical motifs of held-out TFs (Fig. 2).

Predicted motifs form confident protein–DNA complexes

A correlation-based benchmark measures agreement with a reference PWM, but it cannot say whether a predicted motif is biophysically sensible, and it inherits any bias in the reference. We therefore sought an orthogonal, model-independent assay of motif quality by asking a question that neither TFScope nor DeepPBS was optimized to answer: does the DNA a model predicts actually form a confident complex with its own TF? For each of the 41 benchmark TFs we folded the protein together with its TFScope-ensemble consensus DNA and, separately, together with its DeepPBS consensus DNA, using Boltz-2⁹ with ColabFold multiple-sequence alignments¹⁰. Because the protein is identical within each pair, the comparison isolates the effect of the predicted DNA on the confidence of the modeled interface (ipTM).

TFScope’s sequence-only motifs formed complexes at least as confident as DeepPBS’s structure-derived motifs, and by a margin that is small per TF but highly consistent across the set (mean ipTM **0.957** versus 0.923; mean difference +0.035; TFScope favored in **32** of 41 TFs; Wilcoxon $p = 1 \times 10^{-5}$; mean complex pLDDT 0.948 versus 0.921; Fig. 5). The effect is not an artifact of motif length, since the per-TF difference in ipTM is if anything weakly anti-correlated with the difference in DNA length (Spearman $\rho = -0.43$). Boltz-2 has no knowledge of either predictor and was not involved in training or scoring, so this result is independent evidence that TFScope’s sequence-only read-out is not merely correlated with reference PWMs but is physically coherent, producing DNA that its cognate TF can bind in a confidently modeled complex.

A frozen language model encodes DNA-recognition signals

The parity results raise a mechanistic question: if the backbone is frozen and the adapter is small, where does the recognition signal come from? The read-out hypothesis makes a testable prediction, namely that the information TFScope needs should be linearly decodable from the frozen backbone before any specificity training. We tested this directly by probing frozen ESM-2 embeddings for two kinds of knowledge that a specificity model would need.

First, a linear probe on frozen residue embeddings predicts which DBD residues contact DNA, defined as residues within 4.5 Å of DNA in co-crystal structures, with a pooled AUROC of **0.95** and AUPRC of 0.78 across 2,324 complexes and 240,072 residues, evaluated with grouping by protein sequence so that no TF is shared between training and test folds (Fig. 6). The backbone thus already knows, from sequence alone, which residues form the recognition interface. Rendered onto a homeodomain–DNA co-crystal (PBX1, PDB 1B72), the residues the probe scores as DNA-contacting are precisely those of the recognition helix that inserts into the major groove and hydrogen-bonds to the bases, matching the true crystallographic contacts (Fig. 6). Second, the same frozen embeddings organize TFs by structural family, with a k-nearest-neighbor purity of 0.83 and a linear family-probe accuracy of 0.96, so that clade identity is also available without supervision. Both signals are present before any TFScope training, which supports the view that the backbone supplies the recognition content and the contact-aware head mainly converts it into a calibrated, position-resolved motif. This also explains two negative results reported below, that a more elaborate biophysical head does not improve on the simple one and that a weaker backbone degrades accuracy: the informative bottleneck is the frozen representation, not the read-out.

TFScope resolves specificity within transcription-factor families

Because the backbone encodes family identity so cleanly, a natural worry is that TFScope succeeds trivially, by recognizing a TF’s family and returning that family’s average motif. The concern is not hypothetical: many members of a structural family share a core motif, so a model that learned only to classify a DBD into its family and emit the family consensus would still post a high aggregate correlation while learning nothing about the individual protein. Aggregate accuracy alone therefore cannot establish that a model reads specificity, and distinguishing genuine within-family resolution from family lookup is essential for a specificity method to be useful on paralogs, on variants, and on designed proteins that fall inside a known fold.

To separate these two contributions we compared TFScope against a controlled baseline that is given the correct family but not the correct member. For each held-out TF this baseline returns a randomly chosen real PWM from the same family, so any advantage TFScope holds over it must come from resolving differences between members rather than from knowing the family. We then decomposed TFScope’s advantage over a fully random real-PWM baseline into the part recovered by the family-only baseline, which measures the value of family identity, and the residual, which measures genuine within-family specificity, and assessed significance by permutation. On the clustered holdout, family identity accounts for 41% of the advantage, but the majority, **59%**, reflects within-family specificity that a family-average model cannot reach (permutation $p = 2 \times 10^{-4}$; Fig. 7). TFScope therefore reads binding preference at the level of the individual protein, resolving paralogs that share a family fold but differ in their fine preference, and the earlier finding that the backbone encodes family identity is a partial explanation of the model’s accuracy rather than the whole of it.

TFScope captures mutational rewiring and designed proteins

Two applications probe whether TFScope generalizes beyond reproducing the motifs of natural TFs. The first is mutational rewiring, in which a single amino-acid change redirects a TF to a different site. This is a demanding test because the two proteins differ by one residue, so a model that keys on family or global sequence identity will predict nearly identical motifs for both. We used a directional switch score that compares the model’s relative preference for two candidate sites before and after the substitution, isolating the direction of the change rather than its absolute magnitude. On the classic MyoD1 example, in which the L122R substitution in the basic region alters E-box recognition, TFScope reproduces the switch with a strongly positive score ($\Delta_{\text{switch}} = +9.17$; Fig. 8), and an independent structure-based ensemble median agrees that the mutant favors the canonical CACGTG site. The model therefore registers the qualitative consequence of a specificity-altering substitution, a capability that is complementary to, and should not be confused with, the quantitative point-mutation sensitivity we examine in the next section.

The second application is prediction for proteins that do not occur in nature. *De novo* designed DBDs are the hardest possible generalization test for a sequence-only model, because they have no homologs anywhere in the training data, carry no family label that a lookup strategy could exploit, and have no measured specificity to leak; they are also exactly the case where a structure-based pipeline pays the most, since a complex must be predicted before specificity can be read. Applied to a panel of designed DBDs, TFScope recovers the intended CAC-type preference to a graded degree, most clearly for the design DBP6, whose predicted motif correlates with the intended target at $r = 0.65$ (Fig. 9); recovery is partial for the remaining designs, consistent with the expectation that novel proteins fall at the edge of the model’s competence. Even partial, sequence-only recovery is useful as a first-pass readout that can triage designs before committing to structure prediction

or experiment. [PLACEHOLDER — expand with the remaining design cases and any experimental validation once finalized]

Point-mutation sensitivity is bounded by the frozen backbone

The same design choice that makes TFScope general also bounds it in one regime, and we characterize this limit transparently rather than leaving it for readers to discover. The most stringent test of quantitative specificity is a matched wild-type and mutant comparison, where the protein changes by a single residue and the only correct answer is the measured change in binding. On the Barrera panel of homeodomain wild-type and mutant pairs¹¹, TFScope is largely insensitive to the mutation: it predicts a mean absolute change in the motif of 0.005 against an observed change of 0.180, roughly thirty-fold too small, and its predicted changes are essentially uncorrelated with the measured ones.

The cause is structural and, we argue, informative rather than incidental. A single substitution moves the frozen ESM-2 representation only slightly, because the backbone was trained to encode families of related sequences and is by design robust to small perturbations, so a read-out built on that representation inherits the same insensitivity and under-responds to individual residue changes even as it captures family- and clade-level preferences well. This explains why the limitation coexists with the successful switch score of the previous section: the switch score reads a coarse, directional preference between two designated sites and tolerates a muted response, whereas the Barrera comparison demands the fine magnitude of each perturbation, which the frozen features do not resolve. The practical consequence is a clear scope statement for users. TFScope is well suited to predicting the specificity of a TF or a designed protein from its sequence, but it should not, in its present form, be used to score the effect of individual coding variants on binding, a task for which mutation-sensitive features are required. Closing this gap without sacrificing the model’s structure-free generality is the central open problem, which we take up in the Discussion.

Discussion

TFScope shows that accurate prediction of TF binding specificity does not require a protein–DNA structure. A frozen protein language model already carries the recognition signal, and a small structure-free adapter is enough to read it out to motif-level accuracy that matches a strong structure-based method under strict leakage control. This result matters less as a leaderboard update than as a change in what is required to obtain a PWM: the large downstream ecosystem that consumes PWMs can now be driven from sequence alone, wherever a DBD sequence is known.

Two features make the parity credible rather than a metric artifact. We controlled TF leakage explicitly, using both a homology-clustered holdout and a gene-disjoint retrain of the competing method, and observed parity under both; leakage is the most common reason that sequence-only comparisons have been unreliable, and removing it did not erase the result. We also validated the predictions with an assay that is independent of both models, folding each predicted motif with its TF and finding that TFScope’s sequence-only DNA forms complexes at least as confident as the structure-derived DNA. Agreement between a reference-based benchmark and a reference-free biophysical assay is stronger evidence than either alone.

The interpretability analyses locate the signal and reframe what the model is doing. The frozen backbone linearly encodes both DNA-contact residues and family identity, and TFScope’s advantage over baselines is majority within-family rather than family lookup, so the model reads specificity at the level of the individual protein while drawing on representations that were never trained for DNA binding. A practical implication is that the backbone, not the adapter, sets

the ceiling. Two negative results support this reading: replacing the simple motif head with a more elaborate biophysical recognition-energy decoder did not improve accuracy, and substituting a weaker backbone reduced it. Effort is therefore better spent on the representation than on the read-out.

The same analysis explains the model’s principal limitation. Because a single substitution barely moves a frozen ESM-2 representation, TFScope under-reads the fine magnitude of specificity-altering point mutations, even as it captures family- and clade-level preferences and the directional consequences of some substitutions. Progress most likely requires backbone features that respond to individual residues, for instance contact- or structure-conditioned representations, obtained without reintroducing a hard structure dependency at inference; supplying such structural information only during training, as our contact-aware correction already does, is one promising direction. **[PLACEHOLDER — position relative to concurrent PLM-based specificity work once the literature search is complete]**

For practitioners, TFScope turns specificity prediction into a single forward pass on a protein sequence, with no structure required at inference. This is most valuable exactly where structure-based methods are weakest, for the many TFs without a co-crystal and for the rapidly growing space of designed DNA-binding proteins, and it positions sequence-only prediction as a default first step whose motifs can then be refined or validated by structure when one is available.

Methods

Problem formulation

Given the amino-acid sequence of a TF DNA-binding domain (DBD), TFScope predicts a position weight matrix $P \in \mathbb{R}^{4 \times L}$ whose columns are probability distributions over the four nucleotides, together with a soft gate $g \in [0, 1]^W$ over a fixed maximum of W candidate positions that selects the informative motif core. No structural information about the protein or the DNA is used at inference; the only input is the DBD sequence.

Backbone and feature extraction

The backbone is ESM-2 650M (esm2.t33.650M.UR50D, 33 transformer layers, 1,280-dimensional residue embeddings)⁷, kept frozen throughout training. Rather than using a single layer, per-residue features are computed as a learned softmax-weighted average over the last four transformer layers, so that the model can combine the more abstract features of the deepest layers with the more local features of shallower ones; the four mixing weights are the only backbone-associated parameters that are learned. Each residue feature is concatenated with a binary indicator marking whether the residue lies inside the annotated DBD, giving the downstream modules an explicit notion of the recognition domain. **[TODO: confirm exact set of averaged layers and indicator dimensionality]**

Adaptation modules

Four lightweight modules operate on the frozen features. (i) Low-rank adaptation (LoRA, rank 16)⁸ injects trainable low-rank updates into the backbone attention projections, the only mechanism by which gradients reach backbone computation. (ii) A token-level residue mixture-of-experts with two shared experts and eight routed experts, gated by a learned router that is conditioned on a TF-family embedding, processes each residue; the shared experts capture family-invariant recognition chemistry while the routed experts specialize to structural classes, avoiding a proliferation of per-family heads. (iii) Dual gated-attention pooling summarizes the refined residues twice, once over all residues and once restricted to DBD residues, and the two pooled vectors are combined into a protein representation that reflects both domain context and the recognition surface. (iv) A position gate head applies a projection head to the pooled representation to

emit the soft position gate g and the representation consumed by the motif head. [TODO: exact LoRA target modules; MoE router type and top- k ; family-embedding dimensionality; gated-attention pooling equations; per-module parameter counts]

Contact-aware PWM head

The motif head produces the PWM in two additive stages, $P = \text{softmax}(Z_{\text{prior}} + Z_{\text{corr}})$, where a prior term Z_{prior} predicts a coarse position-wise motif from the protein representation and a correction term Z_{corr} refines it using a DNA-contact signal read from the backbone. During training, the contact pathway is supervised with recognition-residue labels, defined as DBD residues within 4.5 Å of DNA in co-crystal structures, which encourages the correction to attend to residues that physically contact DNA. This supervision is applied during training only; at inference the correction is computed from sequence, so the complete model remains structure-free. [TODO: exact correction parameterization (bias vs. full logits), contact-loss weight, and how contact labels are aligned to DBD positions]

Training objective and optimization

The model is trained to match the reference PWM together with the correct motif span. The loss combines a PWM term that compares the predicted and reference columns over the gated positions with a gate term that supervises the soft position gate, and, during training only, the recognition-residue contact term described above. Because a predicted motif can be committed in an arbitrary register relative to the reference, the PWM term is evaluated after aligning the predicted core to the reference. Only the adaptation modules and the layer-mixing weights are updated; the ESM-2 parameters remain frozen. [TODO: exact loss weights, optimizer, learning rate and schedule, batch size, number of epochs, early-stopping/checkpoint-selection criterion, hardware]

Training data, DBD cropping, and leakage-controlled splits

Training pairs associate a TF DBD sequence with a reference PWM (`tf_pwm_training_v23`). Inputs are cropped to the DBD so that the model learns to read the recognition domain rather than incidental flanking sequence; feeding full-length proteins is out of distribution and degrades predictions, so DBD cropping is applied consistently at train and test time. Two evaluation splits are used. The primary benchmark is a held-out set clustered at 40% DBD sequence identity, so that no test protein is a close homolog of any training protein; this is the split on which the 84-structure, 41-TF comparison is reported. For the direct DeepPBS head-to-head we additionally construct a gene-disjoint set of 20 TFs and retrain DeepPBS with those genes removed, so that neither method has seen the test proteins. [TODO: primary data sources (JASPAR/HT-SELEX/CIS-BP), per-family counts, PWM normalization and pseudo-counts, deduplication procedure, exact train/validation/test sizes, clustering tool and threshold details]

Ensembling by register alignment

The reported model is a five-seed ensemble (seeds 42, 1, 7, 13, 23). Each member commits its own motif core through the position gate, and because members may commit in different registers, they cannot be averaged directly. We align each member’s core to the seed-42 core by searching over offsets (± 10 positions) and the reverse complement with a minimum overlap of three positions, then average the aligned PWMs and renormalize each column. The same register-alignment procedure defines the ensemble consensus DNA used in the foldability assay.

Evaluation metrics

PWMs from different methods do not share a coordinate frame, so all comparisons use register-invariant scoring. The primary metric (Panel A) is an oracle-aligned motif *content* correlation: the predicted motif is aligned to the reference by offset and reverse complement to maximize overlap, and Pearson r is computed over the aligned informative columns. This measures the fidelity of the motif itself, independent of where

the model placed it. A secondary end-to-end metric (Panel B) is a coverage-aware correlation (covR) that additionally penalizes gate errors by scoring over the model-committed span, capturing whether the model both predicts the right motif and localizes it. For the DeepPBS comparison we also report a mean gate-oracle- r protocol in which each method’s information-content core is oracle-aligned to the reference before scoring, so that both methods are evaluated on equal footing. **[TODO: exact equations for content- r , covR, and the gate-oracle protocol; definition of informative-column selection]**

DeepPBS comparison

DeepPBS⁵ was run as a five-model ensemble with its published `prot_shape` protein-only input configuration, in a dedicated software environment matching the method’s requirements. Two comparisons are reported. In the leakage-controlled comparison, DeepPBS is retrained from scratch after removing the 20 benchmark genes from its training set, so that its predictions on those genes reflect generalization rather than recall. For reference we also note the value obtained from the publicly released pretrained checkpoints, but because those checkpoints were trained on all 20 test genes we treat that number as leakage-contaminated and do not use it for the head-to-head claim. Both methods are scored with the same register-invariant protocol described above. **[TODO: exact DeepPBS commit/version, torch and PyTorch-Geometric versions, retraining hyperparameters, structure inputs used for the benchmark TFs]**

Independent foldability assay

For each of 41 TFs we constructed two protein–DNA complexes, one with the TF and its TFScope-ensemble consensus DNA and one with the same TF and its DeepPBS consensus DNA, and folded both with Boltz-2⁹. Each DNA was modeled as a double-stranded duplex (the consensus and its reverse complement), and multiple-sequence alignments were obtained from the ColabFold server¹⁰. Because the protein is identical within a pair, its alignment is reused across the two conditions, and the comparison isolates the effect of the predicted DNA. From each prediction we read the interface confidence (ipTM) and the complex pLDDT from the Boltz confidence output, and paired the two conditions by TF. We assessed the difference with a Wilcoxon signed-rank test over the 41 pairs and controlled for motif length by computing the Spearman correlation between the per-TF difference in ipTM and the difference in DNA length. Only the 28 TFs whose ensemble consensus differed from the single-seed consensus were re-folded for the ensemble; the remaining folds and all DeepPBS folds were reused unchanged.

Contact-residue and family probes

To test what the frozen backbone encodes, we probed ESM-2 residue embeddings without any TFScope training. For the DNA-contact probe, residues were labeled as contacting if any atom lies within 4.5 Å of DNA in the co-crystal structure, and a linear classifier was trained on frozen layer-33 embeddings and evaluated with GroupKFold grouped by protein sequence, so that no TF appears in both training and test folds; the probe spans 2,324 complexes and 240,072 DBD residues. We report pooled AUROC and AUPRC across folds. For the family analysis, we summarized each DBD by its frozen embedding and measured k-nearest-neighbor label purity and the accuracy of a linear family classifier. For visualization, the per-residue contact-probe score was written into the B-factor column of PDB 1B72 and rendered in PyMOL, with true crystallographic contacts highlighted.

Within-family decomposition

To separate family identity from within-family specificity we compared TFScope against two baselines on the clustered holdout. A fully random baseline returns a randomly chosen real PWM from the corpus; a family-only baseline returns a randomly chosen real PWM from the same family as the target. TFScope’s advantage over the random baseline measures its total signal, and the family-only baseline’s advantage over the random baseline measures the portion attributable to family identity alone; the difference is the within-family contribution. We repeated the random draws to form null distributions and assessed significance by permutation. **[TODO: number of random draws/permutations and exact baseline PWM sampling scheme]**

Mutation sensitivity and directional switch score

Quantitative mutation sensitivity was evaluated on wild-type and mutant homeodomain pairs from Barrera et al.¹¹. For each pair we computed the mean absolute change between the predicted wild-type and mutant PWMs and compared it with the measured change in specificity, reporting both the average magnitude and the correlation between predicted and observed changes. Separately, the directional switch score evaluates whether a substitution shifts preference toward the expected site: it compares the model’s relative preference between two designated candidate sites for the wild type and the mutant, so that a positive score indicates the predicted shift agrees in direction with the known biology. [TODO: exact Δ_{switch} formula and the number of Barrera pairs used]

Statistics

Paired comparisons between methods use the Wilcoxon signed-rank test; the benchmark head-to-head difference uses a paired bootstrap over TFs; the within-family decomposition uses a permutation test. All tests are two-sided. [TODO: sample sizes per test, bootstrap resample count and CI method, and any multiple-comparison handling]

Code and data availability

[TODO: repository URL, model checkpoints (DOI), processed training/benchmark data deposition, and the DeepPBS retraining recipe]

Declarations

Author contributions (CRediT). [TODO: fill per author]

Competing interests. The authors declare no competing interests. [TODO: confirm]

Funding. [TODO: grants]

Data availability. [TODO: deposition + accession]

Code availability. [TODO: repo + license]

AI-use statement. [TODO: per Nature Methods policy; drafting assistance used]

Figures

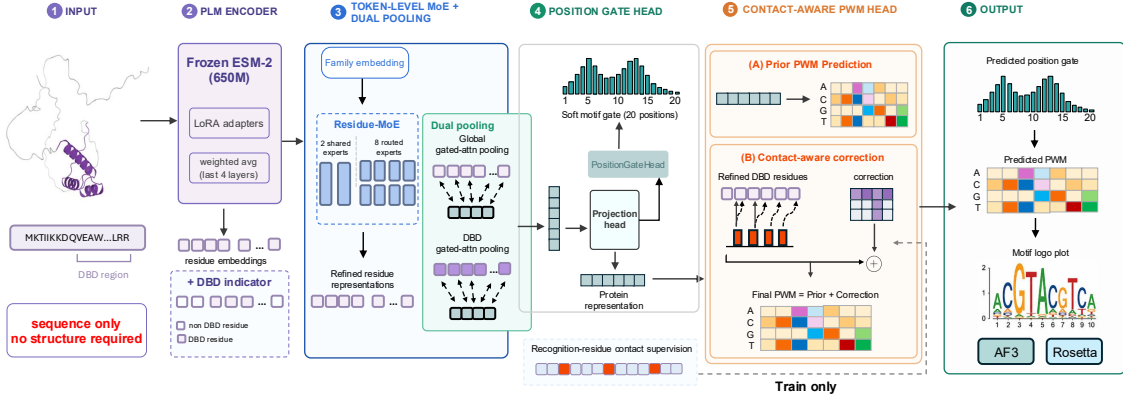


Figure 1: TFScope architecture (sequence-only, no structure at inference). (1) Input: a TF DNA-binding-domain (DBD) protein sequence. (2) PLM encoder: a *frozen* ESM-2 650M backbone with LoRA adapters; residue embeddings are a weighted average of the last four layers, concatenated with a DBD indicator. (3) Token-level residue mixture-of-experts (2 shared + 8 routed experts) conditioned on a family embedding, followed by dual gated-attention pooling (global and DBD-restricted). (4) Position gate head: a projection head produces a soft motif gate over positions and a protein representation. (5) Contact-aware PWM head: a prior PWM (A) is refined by a contact-aware correction (B) driven by recognition-residue contact supervision (train-only), giving Final PWM = Prior + Correction. (6) Output: the predicted position gate, PWM, and motif logo; predicted consensus DNA can be validated by downstream folding (AF3/Boltz) or Rosetta.

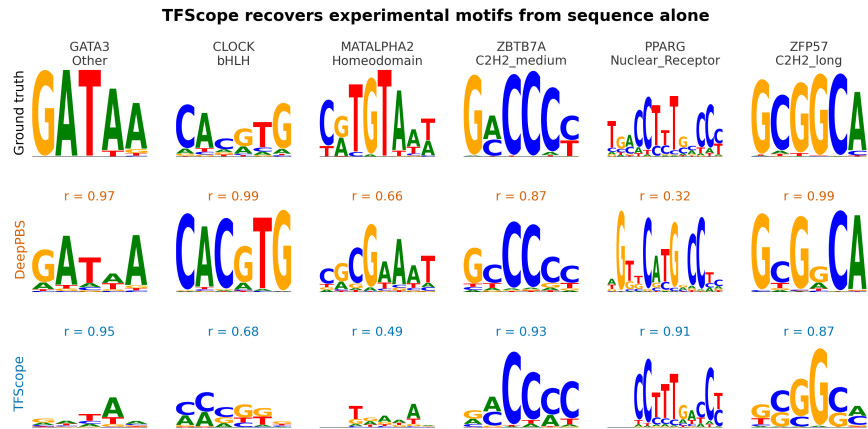


Figure 2: Representative predicted motifs. TFScope-ensemble predicted logos for held-out TFs.

v24-ensemble: benchmark recovery by family (deeppbs_cluster40 test, 84 samples / 41 TFs)

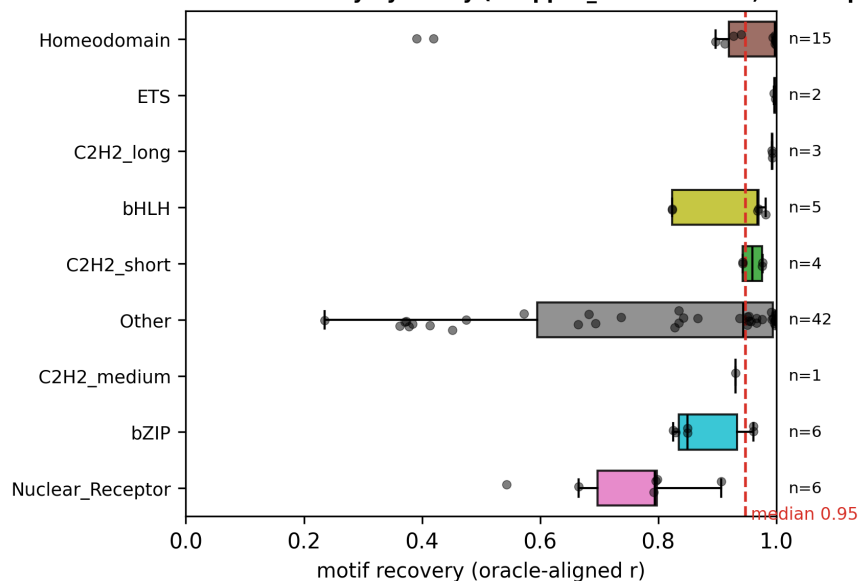


Figure 3: Leakage-controlled benchmark, per family. Motif content correlation on the clustered holdout ($n=84$, 41 TFs); ensemble 0.664. Head-to-head with DeepPBS is a statistical tie (mean gate-oracle r 0.657 vs 0.626; bootstrap $p = 0.30$).

[PLACEHOLDER — benchmark bar + ablation ladder panel]

Figure 4: Benchmark and ablation ladder. [TODO: assemble from results/iclr_phase1_apples_to_apples].

v24 ensemble: sequence-only consensus vs structure-based DeepPBS, judged by Boltz-2 foldability

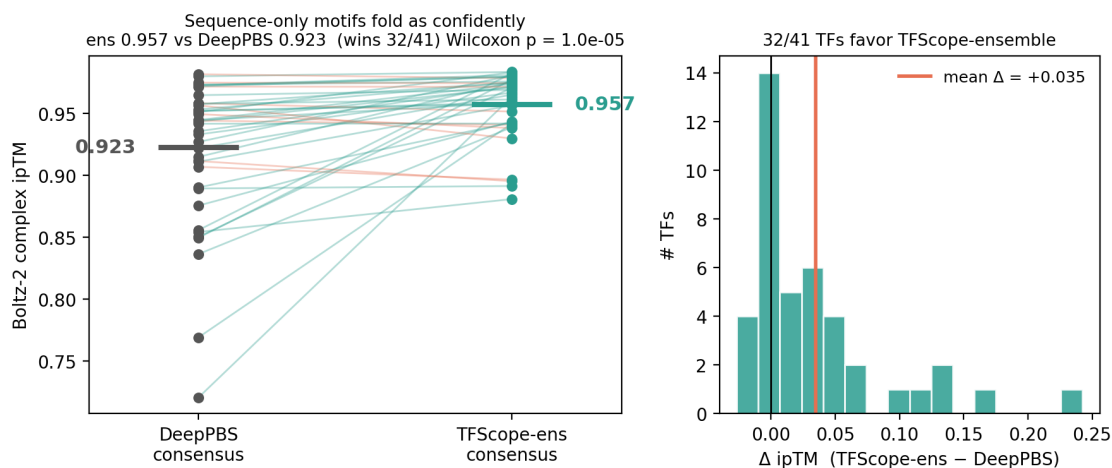


Figure 5: Independent foldability. Boltz-2 complex ipTM for TFScope-ensemble vs DeepPBS consensus DNA (41 TFs): 0.957 vs 0.923, 32/41 favor TFScope, Wilcoxon $p = 1 \times 10^{-5}$.

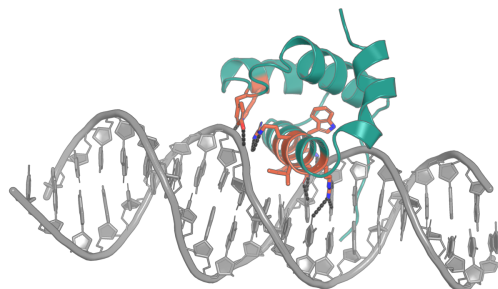


Figure 6: DNA-contact residues recognized by TFScope. PBX1 homeodomain (PDB 1B72): recognition-helix residues scored as DNA-contacting by the frozen- backbone contact probe (AUROC 0.95) shown as orange sticks with H-bonds to the major-groove bases.

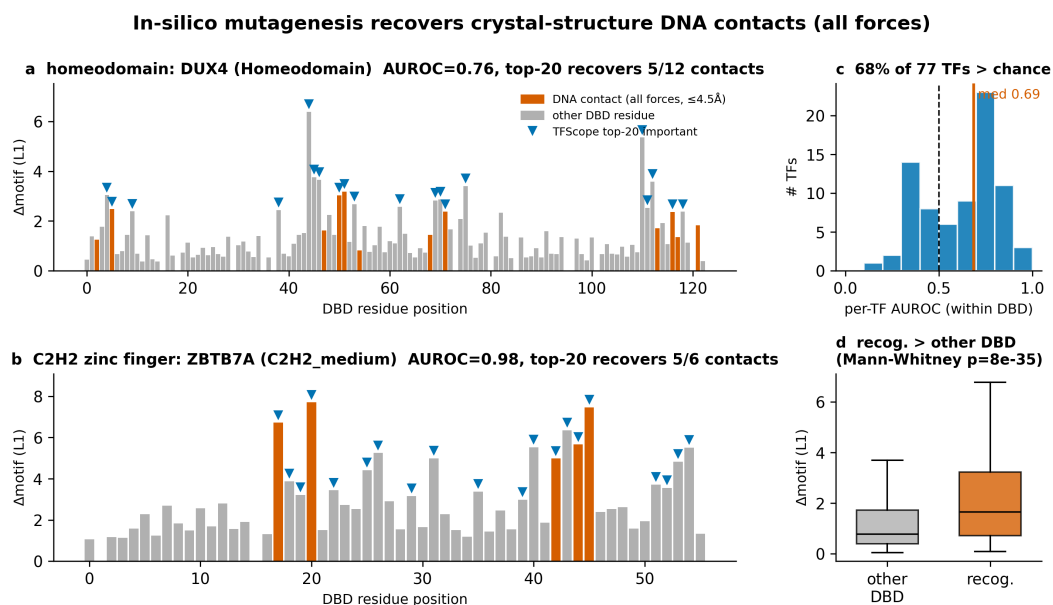


Figure 7: In-silico mutagenesis / within-family signal. [TODO: finalize caption + add decomposition panel].

L122R(v24) shifts MyoD1's predicted preference toward the MYC-like E-box (CACGTG)

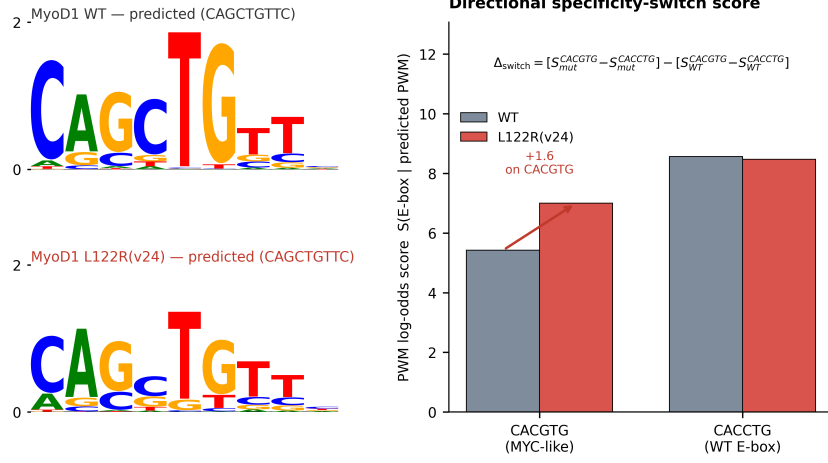


Figure 8: Single-mutation motif switch. MyoD1 L122R E-box switch, $\Delta_{\text{switch}} = +9.17$.

TFScope predicted logo vs experimental mutation-effect heatmap ($\tau=1.0$)

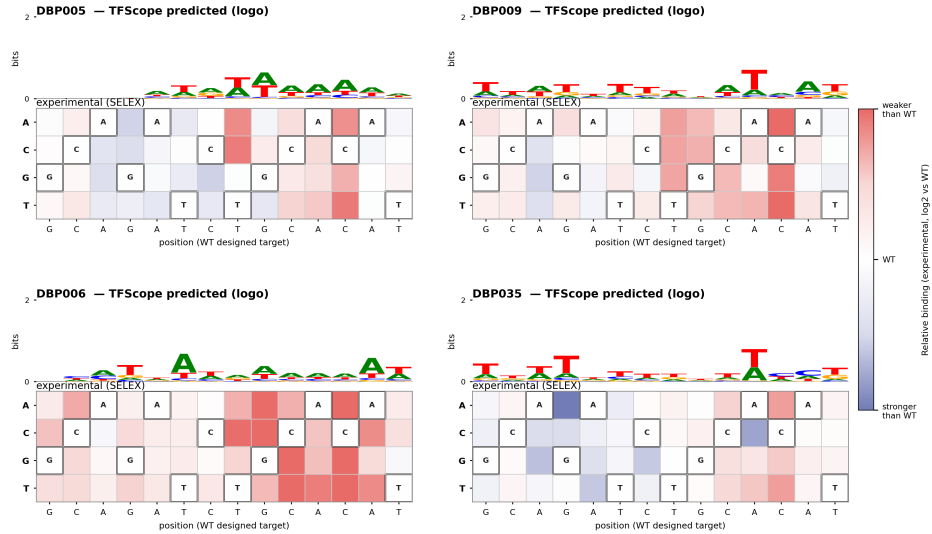


Figure 9: Designed DNA-binding proteins. TFScope recovers intended CAC- type preference (strongest DBP6, $r = 0.65$).

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